ps of dilute $NaHCO_3$ solution. A blue color will
ding the solution may change color to green and

Specie

(1)

ites. On

Sample no - US - 14

huric
) or

1. Physical state: solid
   a) Colour : Brown
   b) Odor : Odorless

c

2. Elementary analysis:

Name of the element	Sulphur (S)	Nitrogen (N)	Halogen (X)	Chlorine (Cl)	Bromine (Br)	Iodine (I)
Test	absent	present	present	present	Absent	Absent

comments: From the elementary analysis, of the sample US-14, I found nitrogen (N) and chlorine (Cl). By treating sodium fusion filtrate with NaOH, $FeSO_4$, $FeCl_3$ the blue colour solution of ferric ferro cyanide $Fe_4[Fe(CN)_6]_3$ is observed. So, we can say that Nitrogen is present. When stock solution is treated with sodium nitroprusside, we observe colour other than the purple or red. So, sulfur is not present in sample US-14.

If we go for halide test, we have to first remove CN^- ion. This is done heating the stock solution with conc. HNO_3. After removing CN^- ion, when we add $AgNO_3$ we can observe a white ppt of the AgX. So, halide (X) is present in US-14.

Now, we should determine the type of halide such as F, Cl, Br, I. To do this we treat stock solution with concentrated Nitric acid (HNO_3) and Carbonate-

tetrachloride (CCl_4). There is no colour in CCl_4 layer. So, Cl^- is present in soln. as we stock solution contains Cl.

3. Solubility test

Solvent	H_2O	5% NaOH	5% $NaHCO_3$	5% HCl	Conc. H_2SO_4	85% H_3PO_4
Solubility	Insoluble	Insoluble	—	Insoluble		

Solubility classification of compounds:

The sample (US-14) is insoluble in H_2O (water), 5% NaOH, and 5% HCl. And the elementary analysis says that the compound containing Nitrogen. So, the compound falls in MN

Class. Miscellaneous Neutral compound.

Class MN: Miscellaneous neutral compounds containing nitrogen or sulphur and having more than five carbon.

Nitrogen present: Anilide, toluidines, amides, imides, nitro acylamines, nitrohydrocarbonyl.

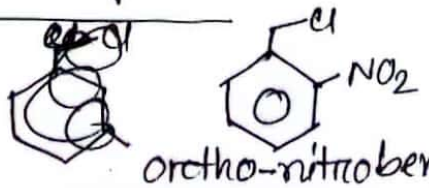
4. Identification of functional group:

Name of functional group	chemical test	Interference
Amides	Nitrous acid test: sample + dil HCl + $NaNO_2$ + shaking	No dorciks effervescence
Anilides	Tafel's test: sample + conc. H_2SO_4 + shaking + $K_2Cr_2O_7$	No green colour on heating

Nitro acyl amine	Reduction of Nitro and then diazotization: Sample + Sn + conc. HCl + boiling + dilute with H_2O. dilute reaction product + conc. HCl + $NaNO_2$ + shaking + sodium Napthoxide	Red orc orange dy was not present.
Nitro acyl amine	Griess reagent test: Sample + covered the test tube with moistered filleterce puper with griss reagent.	A pink colorc on the filter puper way observed
Dinitro and trinitro benzene	Special test: Sample + acetone + 5% NaOH + shaking	Purple colourc way produce

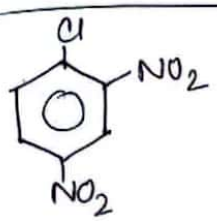
Comments: From the functional group analysis only Nitro group test is positive ~~bay~~ by griss reagent. Furthurc special test of dinitro test is positive. So, compound contains di-nitro as a functional group

5. Melting point: 52–54°C
6. Mixed melting point
7. Litercaturc search: 2,4 dinitrochlorcobenzene (53)
8. Probable compound



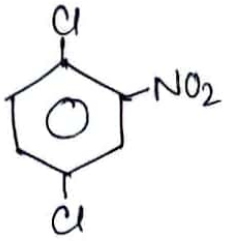
m.p = 49°C

ortho-nitrobenzylchlorcoide



m.p = 53°C

2,4-dinitrochlorobenzene

③ 

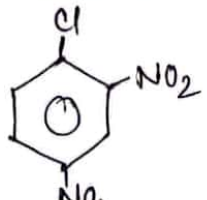
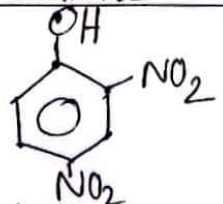
m.p = 55°C

1,4- dichloro-2-nitrobenzene

Comments:

The functional group is dinitro group. So, compound ①, ③ doesn't contains dinitro. So, our sample should be 2.

9. Derivatives:

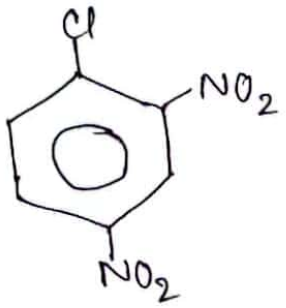
Name of the compound	Derivatives	Melting point
2,4-dinitrochloro benzene	 1-chloro-2,4-diaminine benzene	91°C
	 2,4-dinitrophenol	114°C

ıd 2.0 drops of dilute $NaHCO_3$ solution. A blue col-
nding the solution may change color to green

ːific test for di

olve ab

10. Name & structure of the compound



2,4- dinitrochlorobenzene

Test

(i) Diph
a small t
test tube i
compounds
on heating t
the color on r

(ii) Test with gr
ube. Cover the t
eat the test tube
[/Griess reagen
tion and 0.10%

eduction of nitr

n of nitro comp
ed and about 10.
eaction subsides
ith about 2.0 ml

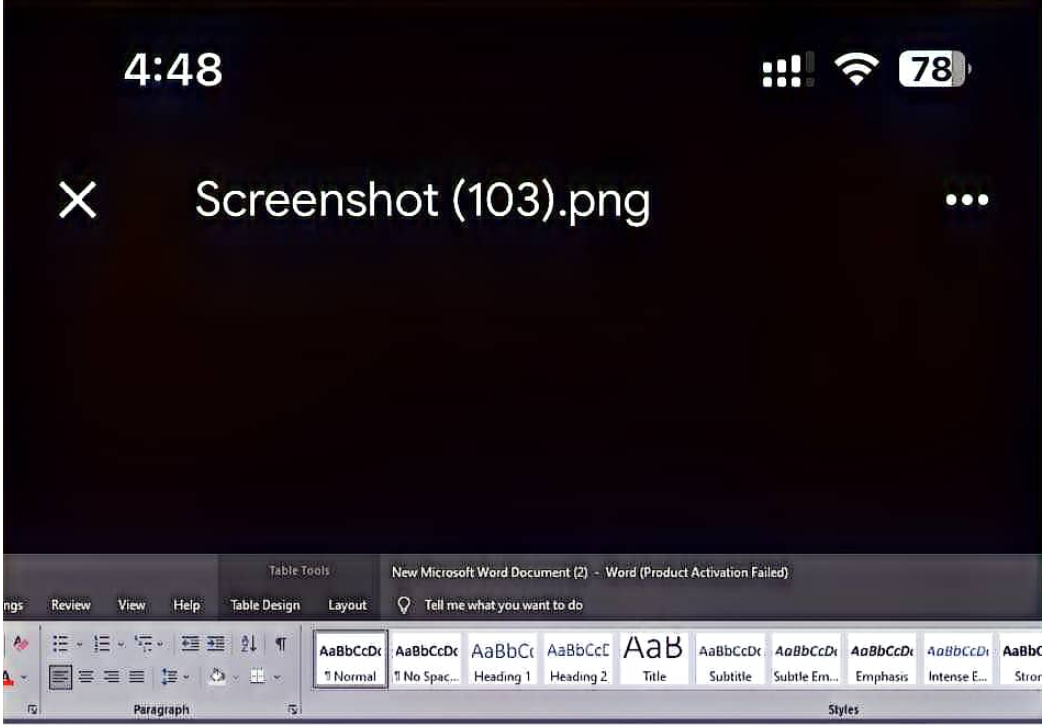
: In a 4.0 inch te
ntrated HCl. Co
of $NaNO_2$ (wh
ise the previous
aphthol in 2.0

Compound	Melting point	Colour
1 chloro-2,4 dinitro benzene	53	yellowlish white
3-Nitrobenzaldehyde	57	Pale yellow
3-Nitrobenzaldehyde	58	Light yellow
4 chloro aniline	72.5	Gray
3-nitroacetophenone	78	White
m-Dinitrobenzene	89.6	Pale yellow
phenanthrene	101	
Acetanilide	114.3	
4-hydroxybenzaldehyde	117	Brownish
Benzoin	132	
2- Chlorobenzoic acid	142	White
2,4-Dihydroxyacetophenone	143	Solid,Powder pink
4-chloro-2-nitrobenzoic acid	144	White
D-glucose	146	White
2-nitrobezoic acid	147.5	
Salicylic acid	158	White
D(+) galactose	167	White
2-Aminophenol	174	Grey
3-Nitroacetophenone	177	white
4-amino phenol	188	
3-Hydroxy Benzoic Acid	201	White
3,5-Dinitrobenzoic acid	206	
pthalamide	238	
3-Hydroxybenzaldehyde	100-103	Brown
O-toluic acid	101-104	White
4-nitro benzaldehyde	103-106	
O- Toluic acid	104 -105	white
4-Amino acetophenone	105-107	
4-hydroxybenzaldehyde	112-116	Brownish
4-Nitro Phenol	113-114	
4-phenyl acetophenone	116-118	
3 amino phenol	120-124	Off white
Benzamide	126-130	White
Benzamide	127-128	White
Succinamide	127-129	White shiny
D-Galactose	128-134	White
2,4 - Dihydroxyacetophenone	143-147	Pink (redish)
2-bromo benzoic acid	147-150	
D(+) galactose	159-161	White yy
4-Bromoacetanilide	164-167	White
4- Bromoacetanilide)	165-169	Off white
4-bromoacetanilide	166-169	Off white
3-Amino benzoic acid	171-174	Brown
Thiourea	177-179	Off white
4-Chloro-3-nitrobenzoic acid	181-184	
4 bromo benzoic acid	252-254	

4-Chloro benzaldehyde	47-48	
4-chloroaniline	69-71	Grey
Benzil	93-96	Light yellow







Name	Melting Point	Colour
3-Nitroaniline	114	Brownish yellow
2-Chlorobenzoic acid	138-140	
Salicylic acid	157-158	White
3-Chloro-p-anisidine	20	
4-Hydroxyacetophenone	145-147	
4-Bromobenzoic acid	252-254	
2,4-Dinitrophenol	108	Pale yellow
4-Nitroacetophenone	80°C	
2,4 dihydroxybenzaldehyde	135-137	Light Red
Benzamide	126-130	
Acetanilide	113-115	
D-Glucose	146	White
1-Chloro-3,4-dinitrobenzene	53	
1,3-Dinitrobenzene	89	
4-Hydroxybenzaldehyde	112-116 °C	
P-chloroaniline	72.5 °C	grey

